Technical specification template: Standby Generator Sets

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A6. Generated 2026-11-08 20:00 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for standby generator sets for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- ISO 8528 series (reciprocating engine driven generating sets)
- IEC 60034 series (rotating electrical machines)
- Local emissions, noise and fuel storage regulations [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit study pending utility, transformer and generator data]	NOT CALCULATED

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Generators per power path	3	PRELIMINARY_ASSUMPTION. N+1: the load is carried with the largest unit out
Rated apparent power (each)	2,000 kVA	UNVERIFIED_PLACEHOLDER. ASSUMED to be a PRP rating. NOT confirmed. ISO 8528: PRP, ESP and COP are different ratings
Rated power factor	0.8	UNVERIFIED_PLACEHOLDER. Engine kW is the kVA rating times this factor
Rating basis required	PRP (prime power) per ISO 8528-1, at the site conditions	Project decision. ESP is not a basis for prolonged outages
Derating (ambient, altitude, installation)	1.0	UNVERIFIED_PLACEHOLDER. 1.00 is a placeholder
Load acceptance and transient limits	[Frequency dip, voltage dip, recovery time: project to specify]	Pending manufacturer data

5. Technical requirements

5.1 The generating set shall be rated for prime power (PRP) at the site conditions; the rating basis shall be stated on the datasheet and the nameplate.

5.2 The manufacturer shall state kW and kVA ratings, the allowed average load factor and any hours restriction for the stated basis.

5.3 The set shall accept the stated load steps, including the UPS rectifier input with its walk-in, within the specified frequency and voltage limits; the manufacturer shall provide transient response data.

5.4 The alternator shall be rated for the harmonic current of the UPS rectifier load [project to specify the harmonic data].

5.5 Minimum-load and light-load operation requirements shall be stated by the manufacturer.

5.6 Control system, paralleling, synchronising, protection, fuel system, cooling, exhaust, starting system and enclosure: [project to define].

5.7 Black-start capability and start-to-load time: [project to define].

6. Vendor data schedule

The tenderer shall complete the schedule below with its offer: the value offered, its unit, and the name and revision of the document that states it. A value without a source document is not accepted. The parameters are those in the manufacturer-data register; supplied values are imported into the calculations through the data override mechanism, and a value stays NOT

VERIFIED until a reviewer records an attestation against its source (name, organisation, date, evidence reference, scope).

Parameter id	Parameter	Datasheet required	Current status	Value offered	Unit	Document and revision
equipment.generator_rating	Generator PRP rating (each)	Generator set data sheet	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
equipment.generator_rating_power_factor	Generator rating power factor	Generator set data sheet	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
equipment.generator_esp_rating	Generator ESP rating (each)	Generator set data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
equipment.generator_cop_rating	Generator COP rating (each)	Generator set data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
derating.generator_ambient_factor	Generator ambient derating factor	Manufacturer derating curve for ambient conditions (generator)	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
derating.generator_altitude_factor	Generator altitude derating factor	Manufacturer derating curve for altitude conditions (generator)	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
derating.generator_installation_factor	Generator installation derating factor	Manufacturer derating curve for installation conditions (generator)	UNVERIFIED_PLACEHOLDER	[vendor to complete]	[vendor to complete]	[vendor to complete]
generators.prp_max_average_load_factor_pct	PRP maximum average load factor	Generator set data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
generators.operating_hours_restriction	PRP operating-hour restriction	Generator set data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
generators.overload_capability	Manufacturer overload capability	Generator set data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
generators.max_step_load_pct	Maximum single load step	Generator set data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
generators.max_frequency_dip_pct	Maximum frequency dip allowed	Generator set data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
generators.max_voltage_dip_pct	Maximum voltage dip allowed	Generator set data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
generators.ups_rectifier_walk_in_s	UPS rectifier walk-in time	UPS data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
generators.u	UPS input	UPS data	MISSING	[vendor to	[vendor to	[vendor to

ps_input_current_thd_pct	current THD	sheet		complete]	complete]	complete]
short_circuit.generator_subtransient_reactance_pct	Generator subtransient reactance x''d	Generator set data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]
short_circuit.generator_x_r_ratio	Generator X/R ratio	Generator set data sheet	MISSING	[vendor to complete]	[vendor to complete]	[vendor to complete]

Model, manufacturer, country of manufacture, and the type-test reports for the offered design: [vendor to complete].

7. Compliance schedule

The tenderer shall state, for every standard in section 2 and every technical requirement in section 5, whether the offer complies. 'Complies' needs an evidence reference (type-test report, certificate, calculation, drawing or data sheet with revision). A statement without evidence is recorded as not demonstrated. The requirement references below are the numbered items of section 5 of this template; no clause of any standard is quoted.

Standard named in section 2	Edition offered	Complies / complies with comment / does not comply	Evidence reference	Comment or deviation
ISO 8528 series (reciprocating engine driven generating sets)	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]
IEC 60034 series (rotating electrical machines)	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]
Local requirements [project to add]	[vendor to complete]	[vendor to complete]	[vendor to complete]	[vendor to complete]

Requirement (section 5)	Complies / complies with comment / does not comply	Evidence reference	Comment or deviation
5.1	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.2	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.3	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.4	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.5	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.6	[vendor to complete]	[vendor to complete]	[vendor to complete]
5.7	[vendor to complete]	[vendor to complete]	[vendor to complete]

8. Inspection and quality requirements

8.1 Review of the set data sheet, alternator and engine data and the paralleling scheme.

8.2 Inspection of the base frame, enclosure and fuel system components.

8.3 Routine tests on every unit as required by the applicable standard and this specification.

8.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

8.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

8.6 Calibration certificates for test instruments.

9. Factory acceptance test (FAT) requirements

9.1 Factory test per ISO 8528 for performance class [project to define]: load runs at agreed steps, regulation, load acceptance and transient response.

9.2 Protection and control function tests; paralleling and load sharing if applicable.

9.3 Witnessing: [owner / independent inspector to attend].

9.4 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9.5 Acceptance criteria are taken from the applicable standard and this specification [project to define]; a test passed against a criterion not stated in advance is not accepted.

9.6 Equipment shall not be shipped until the FAT report is accepted or a concession is recorded and approved by the owner.

10. Site acceptance test (SAT) requirements

10.1 Inspection of mountings, fuel and cooling connections; start tests; a load-bank test at the site conditions for the duration and steps in the approved procedure; control and protection function tests; paralleling and load-sharing tests where specified.

10.2 A SAT procedure shall be submitted for approval before the tests. It states the test, the method, the instrument, the acceptance criterion and the witness, with the criteria taken from the applicable standard and this specification [project to define].

10.3 SAT is carried out on the installed equipment with its final connections, after the installation inspection and before commissioning loads are applied.

10.4 Results are recorded on signed sheets. Any deviation, retest and concession is listed with its disposition. A SAT is not passed while a deviation is open.

10.5 SAT results are compared with the FAT results; a significant difference is investigated and recorded.

11. Commissioning requirements

11.1 Pre-start checks, fuel and coolant, protection settings.

11.2 Load bank test to rated load; transient tests with the real UPS load where possible.

11.3 Utility-failure transfer and return tests with the switchgear; fuel runtime verification against the design basis.

11.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

11.5 Integrated systems testing of the whole power chain (utility loss, generator start and transfer, UPS on battery, loss of one power path, return to normal) is carried out under a project procedure [project to define]. This equipment's commissioning is not complete until it has taken part in that test.

12. Submittals and documentation

Document	When required	Status
Completed vendor data schedule (section 6) and compliance schedule (section 7)	With the tender	[vendor to complete]
Datasheets and general arrangement drawings	With the tender, final before manufacture	[vendor to complete]
Type-test reports and certificates for the offered design	With the tender	[vendor to complete]
Inspection and test plan (ITP)	Before manufacture	[vendor to complete]
FAT procedure and report	Before and after the FAT	[vendor to complete]
Installation instructions and SAT procedure	Before delivery	[vendor to complete]
Operation and maintenance manuals, as-built drawings and final data	Before commissioning; as-built at handover	[vendor to complete]
Recommended spare parts and special tools lists	With the tender and at handover	[vendor to complete]
Training plan and materials [project to define]	Before commissioning	[vendor to complete]
Warranty terms [project to define]	With the tender	[vendor to complete]

13. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- The vendor data schedule (section 6) and the compliance schedule (section 7) are blank until a tenderer completes them.
- FAT, SAT and commissioning acceptance criteria are [project to define].
- This template has not been reviewed by an independent engineer.